

Map a Thermal Gradient

Put identities in places before comparing values

E0 SYNTHETIC REPLAY — NO PROBE, DISPLAY, OR STORAGE CLAIM

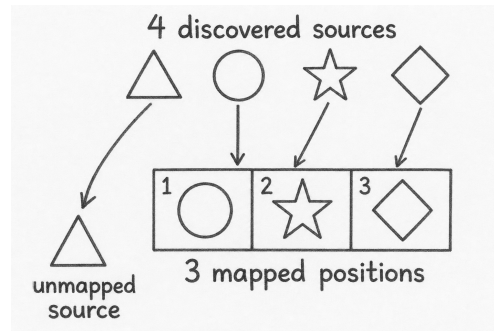
Lesson 066 — project
Prerequisite Lesson 065 qualified snapshot
Input copied snapshot and controls

Time 180–240 minutes
Board compile-only Mega replay
Observation named result cells

Responsibility. `ThermalGradientMapper` projects two to four configured identities into fixed spatial order, derives conservative adjacent intervals, and emits inert page and caller-owned record intent. It does not poll a bus, drive a display, read a clock, write storage, or establish a physical temperature gradient.

1 Give each identity one place

Lesson 065 publishes four stable identified slots. Lesson 066 chooses an ordered subset of two through four of those identities. This list means left-to-right or near-to-far because configuration says so. Search order, arrival order, ROM number, current temperature, and page order never become spatial evidence.



Alternative text: four discovered graphite source tokens are projected by arrows into three numbered mapped positions in configured order; the triangle source remains explicitly unmapped.

Every mapped ROM must occur exactly once among the four source ROMs. One mapped ROM, a repeated ROM, an all-zero ROM, or an unknown ROM rejects configuration. A configured but unmapped source remains permitted; it does not silently become an extra page or gradient.

E0 establishes	E0 does not establish
identity-based spatial projection	package location or physical mounting
deterministic interval arithmetic	calibrated accuracy or simultaneity
inert LCD/LED content intent	a powered display or observation circuit
volatile caller-owned record intent	RTC time, media, or durability

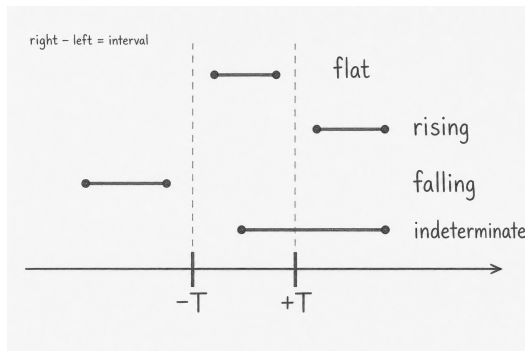
Predict the ordered pair when source enumeration is reversed but configuration is unchanged: _____. Which order has authority? _____.

2 Compare complete intervals

Each current Lesson 065 probe carries a closed signed raw-sixteenth interval. For adjacent configured slots L and R , widened arithmetic computes:

$$[G_{\text{low}}, G_{\text{high}}] = [R_{\text{low}} - L_{\text{high}}, R_{\text{high}} - L_{\text{low}}].$$

The subtraction uses 32-bit fields even though each endpoint is 16-bit. This prevents an extreme pair from overflowing before classification.



Alternative text: four graphite gradient intervals sit around negative and positive threshold marks on one number line, showing complete interval examples classified as flat, rising, falling, and indeterminate.

For a positive threshold T , **Rising** requires $G_{\text{low}} \geq T$, while **Falling** requires $G_{\text{high}} \leq -T$. **Flat** requires the complete interval to remain strictly between the two boundaries. An interval that crosses a boundary is **Indeterminate**; the mapper never rounds it into a direction.

Gradient interval	Threshold $T = 16$	Quality
$[16, 20]$	lower equals T	Rising
$[-20, -16]$	upper equals $-T$	Falling
$[-15, 15]$	wholly inside	Flat
$[15, 17]$	crosses T	Indeterminate

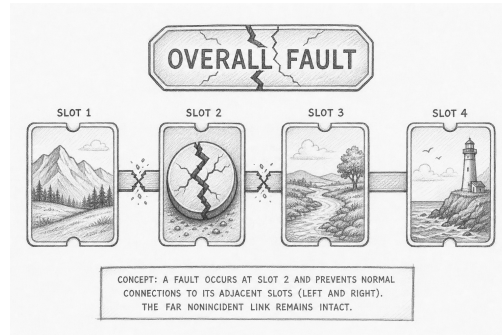
3 Aggregate without hiding disagreement

Any mapped probe or pair fault makes overall health **Fault**. Otherwise, an indeterminate pair or a mixture of rising and falling pairs makes **Disagreement**. At least one directional pair makes **Gradient**; all-flat pairs make **Stable**.

Equal extrema use the lowest mapped slot for primary attribution while tie masks preserve all equal slots. Calculate the interval for $L = [400, 407]$, $R = [416, 419]$: _____ through _____.

4 Let faults dominate every affected page

A probe contributes a number only while its copied quality is **Current** and its inclusive **freshThrough** bound admits the mapper's supplied **now**. Equality remains fresh; the next representable millisecond is stale. Missing, pending, CRC-failed, resolution-mismatched, stale, implausible-step, and transport-fault evidence must never appear as zero, "cold," flat, or cached current data. The mapper coherence-checks the copied input age, then recomputes every emitted page age and record age from that probe's **observedAt** to the mapper envelope's **now**. A control edge cannot refresh either time.



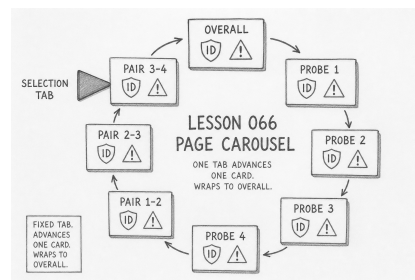
Alternative text: four graphite slot cards are joined by three connector bars; slot two and both incident bars are cracked beneath one global overall-fault banner, while the nonincident bar between slots three and four remains intact.

A bad interior slot faults both incident gradients. A healthy nonincident pair may retain its own numeric evidence, but every page carries the complete overall fault mask. Page selection and record controls cannot alter that classification.

Page	Healthy content	Fault content
Overall	health, extrema, complete mask	health and complete mask
Probe	ROM, interval, age, quality	ROM, age, quality, fault token
Adjacent	both ROMs, interval, direction	both ROMs and fault token

5 Keep pages inert

Page order is fixed: overall, mapped probes in spatial order, then adjacent pairs. A fresh **nextEdge** advances once and wraps from the final page. The intent includes mutually exclusive selectors: **selectedSlot** is meaningful only on a probe page, while **selectedGradient** is meaningful only on an adjacent-gradient page; both are canonical zero on the overall page, and the inapplicable selector is zero on either detail page. LCD content choices, an active-slot LED mask, and grayscale-safe health/fault meaning are values, not hardware operations.

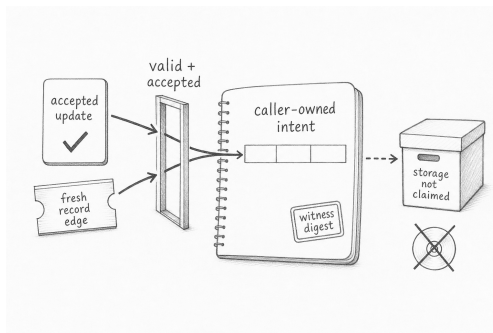


Alternative text: graphite cards follow a fixed loop from the overall page through configured probe pages and adjacent-pair pages, with one copied edge moving a pointer exactly once and wrapping to the beginning.

A control has its own owner, source, configuration revision, sequence, time, and status. Changed duplicates, regressions, future controls, half-range ambiguity, stale controls, and exhaustion reject atomically.

6 Create record intent, not a stored record

A fresh `recordEdge` may fill exactly one bounded caller-owned `ThermalMapperRecordImage`. The edge can capture a changed or unchanged accepted probe frame. Record creation belongs to the control edge, not to a temperature change.



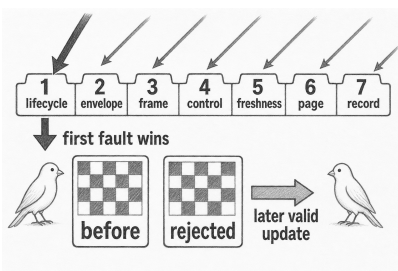
Alternative text: accepted update and fresh record-edge cards cross a validation barrier into a caller-owned notebook with a witness digest; a separate crossed-out storage claim remains outside the owned result.

The record preserves owner and lifecycle, configuration revision, nonzero record sequence, complete edge attribution, source set identity and cycle, mapping time, every mapped ROM and interval-or-fault, every adjacent gradient, overall health and mask, all four configured source ROMs including permitted unmapped identities, and a deterministic digest covering that complete witness. Full fields remain authoritative; the digest is not a substitute for them.

Volatile boundary. `hasRecord` means “the caller received one record intent in this synchronous update.” It does not mean a file was opened, bytes were stored, a clock was accurate, media acknowledged a write, or data survived reset or power loss.

7 Make the whole update atomic

The mapper validates the complete copied envelope before committing policy or touching the caller’s result. A rejected update preserves every policy byte and every result canary. A byte-identical duplicate is idempotent: it cannot advance a page, emit a second record, refresh evidence, or extend freshness. An exact envelope replay may republish the identical current intent image to the caller, but it creates no new record and consumes no record sequence.



Alternative text: seven large graphite stages run from lifecycle through envelope, frame, control, freshness, page, and record; first fault wins, a rejected checkerboard remains identical to the before image, and a later valid update proceeds between two canary birds.

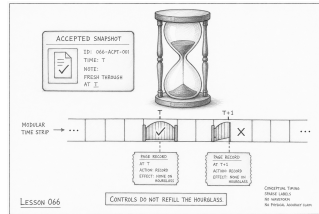
Record-edge and record-sequence exhaustion fault before zero. Reset starts a new lifecycle and clears page and record history. Shutdown makes outputs inactive and is idempotent; reinitialization must accept a new complete frame.

8 Replay the project without touching hardware

The canonical Mega sketch constructs copied Lesson 065 evidence, maps its identities, supplies copied page and record edges, and places selected results in named volatile cells. The sketch follows acquire, configure, start, then observe, decide, actuate—where actuate means publish inert intent.

1. **Predict:** write mapped order, three adjacent qualities, overall health, page index, and whether one record intent should appear.
2. **Observe:** compile for Mega 2560 and inspect the named volatile result cells in a debugger, simulator, or disassembly. Serial may support the observation but is not required.
3. **Interpret:** compare identities, intervals, fault mask, page, record edge, and sequence with the copied fixture. Explain any disagreement before changing evidence.

This is the E0 non-Serial observation path. It establishes deterministic software acquisition and publication separately from an electrical safe-state experiment. E0 owns no endpoint, so powered resource acquisition, released-line state, display behavior, and probe behavior remain future E1 checks.



Alternative text: an accepted snapshot is fresh through time T beside an hourglass and modular time strip; record controls at T and $T + 1$ do not refill the hourglass, so T is fresh; at $T + 1$ the probe is classified stale and contributes no number.

9 Publisher verification and resource boundary

Host evidence	Strict tests, sanitizer tests, standalone header, style, and complete atomic/canary replay are publisher gates.
Mega evidence	Ordinary and isolated no-LTO flash/static SRAM measurements are inserted from the canonical replay before promotion.
Memory evidence	Mapper object, result, record image, hidden-return behavior, conservative synchronous stack, recurring Lessons 064–066 placement, and residual SRAM are independently reproduced before promotion.
Document evidence	PDF policy, metadata, extraction, monochrome, page-count, and rendered-pencil review are publisher-owned gates.
Physical evidence	Deferred; no powered probe, display, RTC, media, thermal accuracy, immersion, heater, or preservation claim is authorized.

Until exact measurements replace this pre-publication statement, the lesson is not promoted. Planning hard limits are 24 KiB flash, 3,072 B static SRAM, 1,024 B synchronous stack, 768 B mapper object, 384 B per fixed caller-owned ABI value, and at least 4 KiB residual SRAM.

10 Draw the boundary around the result

There is no wiring diagram or formal schematic at E0. A future powered boundary must first qualify exact probe specimens and supply mode, bus loading and pull-up, exact LCD/LED endpoints, pin and resource ownership, electrical safe state, independent observation, and measured acceptance. Any RTC or media work separately needs exact devices, ownership, bounded transactions, receipts, corruption and power-loss recovery, capacity, wear, and durability evidence.

No version of this lesson authorizes immersion, an unknown “waterproof” assembly, a hot surface, heater, flame, ignition, food or medical judgment, preservation control, or unattended thermal stimulus.

11 Exercises and acceptance reasoning

1. Scramble all discovery/input permutations for three identities. Show why the complete mapper image remains in configured spatial order.
2. Test interval boundaries immediately below, at, and above both thresholds. Explain every indeterminate result without rounding.
3. Make the middle probe stale. Mark both incident fault gradients and the healthy nonincident evidence that remains attributable.
4. Advance from the last page to overall. Replay the identical control, then change one copied control field at the same sequence.
5. Emit a fresh record edge over an unchanged qualified snapshot. List what the record proves and five durability claims it cannot prove.
6. Replay ordinary rollover, future time, exact half-range ambiguity, record exhaustion, reset, shutdown, and reinitialize. Confirm every rejected caller image remains byte-for-byte unchanged.

Project result. You have composed copied transport and qualification evidence into a stable, fault-dominant spatial view without smuggling discovery order, hardware control, calibrated accuracy, or durable storage into the policy.

Stop at E0: exact powered specimens, wiring, presentation, recording, thermal accuracy, and every physical acceptance result remain open work.